THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An aircraft ran out of fuel because of a mix-up of units, showing that: 1

(a) planes need no fuel
(b) fuel is not measurable
(c) units must be used carefully and consistently
(d) units do not matter

2. Models of the atom have changed over time because: 1

(a) the atom changed
(b) new evidence required changes
(c) old models were hidden
(d) scientists were bored

3. When a prediction does not match the observation, scientists should: 1

(a) re-examine the model, assumptions or measurements
(b) stop the study
(c) change the observation
(d) ignore the observation

4. A student ignoring air resistance while studying a falling ball is: 1

(a) making a mistake
(b) changing the question
(c) breaking a law
(d) making an assumption to simplify the model

5. Weather forecasts for many days ahead are less reliable because: 1

(a) clouds are invisible
(b) satellites fail
(c) small differences grow in complex systems
(d) weather has no rules

6. Using agreed international units in science helps because: 1

(a) everyone can compare and repeat measurements
(b) it makes numbers smaller
(c) it avoids measuring
(d) it removes errors completely

7. Which is the best example of a scientific prediction? 1

(a) 'Pendulums are old.'
(b) 'It may or may not swing.'
(c) 'I like pendulums.'
(d) 'If I increase the length of a pendulum its time period will increase.'

8. The mass of a body is 2.5 kg. In grams this is: 1

(a) 2500 g **(b)** 250 g **(c)** 25000 g **(d)** 25 g

9. The SI unit of mass is the: 1

(a) kilogram **(b)** tonne **(c)** pound **(d)** gram

10. Chemistry is the study of: 1

(a) only forces
(b) only stars
(c) only living things
(d) the composition, properties and transformations of matter

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the importance of estimation in science? Give an example. 2

12. Why do scientists use mathematics? 2

13. Why is it important to use standard SI units? 2

14. Write the SI units of length, mass and time. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Distinguish between a hypothesis, a law and a theory with one line for each. 3

16. Explain why weather forecasts far into the future are less reliable. 3

17. Why are the idealisations like frictionless surfaces or point masses useful? 3

OR

Why do we need agreed symbols and units in science? Give two examples. 3

18. How can Meghna test Varsha's prediction of rain? Give three steps. 3

OR

State three skills a young scientist should develop. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the importance of estimation with two worked examples and state the situations where exact values are needed. 5

OR

Explain how scientific ideas are shared and checked by other scientists and why honesty is important. 5

20. Describe the SI system: base units, derived units and prefixes, with conversions among km, m, cm and mm. 5

OR

A student predicts that a heavier ball falls faster. Describe how she can test it, what she may find and how the model with air resistance explains it. 5

END OF QUESTION PAPER — Total: 40 Marks